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Late Cretaceous alkaline complexes, southeastern Brazil: Paleomagnetism and geochronology

C.R. Montes-Lauar [a], I.G. Pacca [b], A.J. Melfi [b], K. Kawashita [c]

[a] *Núcleo de Pesquisa em Geoquímica e Geofisica da Litosfera da Universidade de São Paulo (NUPEGEL), Caixa Postal 9638, 01065-970 São Paulo, Brazil*

[b] *Departamento de Geofísica, Instituto Astronômico e Geofísico da Universidade de São Paulo and NUPEGEL, Caixa Postal 9638, 01065-970 São Paulo, Brazil*

[c] *Centro de Pesquisas Geocronológicas, Instituto de Geociências da Universidade de São Paulo, Caixa Postal 20899, 01498-970 São Paulo, Brazil*

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Abstract

Late Cretaceous alkaline rocks from Poços de Caldas, Itatiaia, Passa Quatro and São Sebastião Island were studied with the main purpose of improving the South American apparent polar wander path (APWP). Rb–Sr age determinations for Passa Quatro and alkaline stocks from São Sebastião Island are around 70 and 81 Ma respectively. The magnetic carriers in the rocks studied were maghemite and highly oxidized titanomagnetites. The paleomagnetic poles obtained in this work are 320.1°E, 83.2°S ($N = 47$, $A_{95} = 2.7°$, $K_{SC} = 2609.8$) for Poços de Caldas, 360.0°E, 79.5°S ($N = 18$, $A_{95} = 5.7°$, $K_{SC} = 607.4$) for Passa Quatro and Itatiaia, and 331.9°E, 79.4°S ($N = 18$, $A_{95} = 4.9°$, $K_{SC} = 862.4$) for São Sebastião Island (the ISLAND set).

1. Introduction

The Late Cretaceous is recognized as the most important period for the emplacement of alkaline complexes in South America, and study of the paleomagnetism of these complexes is fundamental to our understanding of their role in the evolution of the South American Platform and for improving the South American APWP for this period. The alkaline magmatism resulted from the tectonomagmatic reactivation (Permian to Tertiary) related to the fragmentation of Gondwana and to the opening of the Atlantic Ocean. These alkaline rocks consist mainly of mildly to strongly undersaturated syenites and nepheline syenites and they are related to tectonic structures such as arches, flexures, rifts and deep faults ([1–4], and references therein).

The alkaline complexes of Poços de Caldas, Itatiaia, Passa Quatro and the stocks from São Sebastião Island (I, II, III and IV in the location map of Fig. 1) that were studied in this work are the largest alkaline bodies known in Brazil. Some of their geological characteristics are presented below. All the K–Ar ages presented in this work have been recalculated using the constants recommended by Steiger and Jäger [5].

2. Geological outline

2.1. Poços de Caldas complex

The Poços de Caldas alkaline complex (800 km^2, Fig. 1) is situated on the boundary between the states

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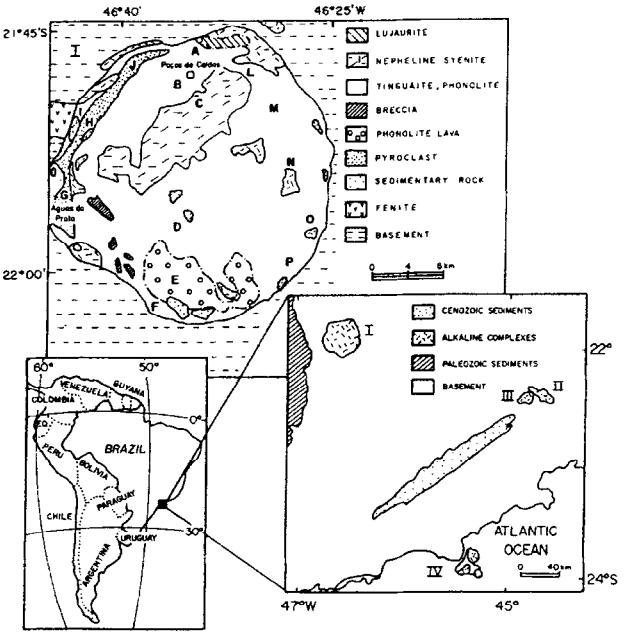


Fig. 1. Geological map of the Poços de Caldas alkaline complex (adapted from [6]) and location map. Letters *A* to *P* represent sampling areas. The inset shows the alkaline complexes: I = Poços de Caldas; II = Itatiaia; III = Passa Quatro; IV = São Sebatião Island.

of São Paulo and Minas Gerais. The complex was intruded into the Late Proterozoic Guaxupé Massif [7,2] and is interpreted as a large caldera structure. Poços de Caldas consists mainly of tinguaites, phonolites and nepheline syenites and were preceded by ankaratritic volcanism [8,9,1,2]. The available

Table 1
Geochronological data for the Poços de Caldas complex

Method	Rock Type	Material	Age (Ma)	R_0	Observation
	Ankaratrites	WR	89	-	-
K-Ar	Tinguaites	WR	83-64	-	-
[10, 11, 12]	Ne-syenites	B, K-felds.	83-61	-	-
	Phonolites	WR	77-87	-	-
	Phonolites	WR	54	-	dyke that cuts regional Ne-syenites.
Rb-Sr [13]	Ne-syenites	WR	89.8±2.8	0.7051±1	regional Ne-syenites.
	Ne-syenites	WR	86.3±6.0	0.7051±1	regional Ne-syenites.
Rb-Sr [14]	Ne-syenites	SP, AE, Ne, K-felds.	77.9±3.1	0.70512±1	internal isochron of regional Ne-syenites.
	Ne-syenites	WR	75.8±1.2	0.70532±11	hydrothermally altered Ne-syenites.
Ar-Ar [14]	Lamproite	B	76±1	-	dyke that crosscuts the hydrothermally altered Ne-syenites.

R_0 = initial $^{87}Sr/^{86}Sr$ ratio; WR = whole rock; Ne = nepheline; B = biotite; SP = sphene; K-felds = K-feldspar; AE = aegirine

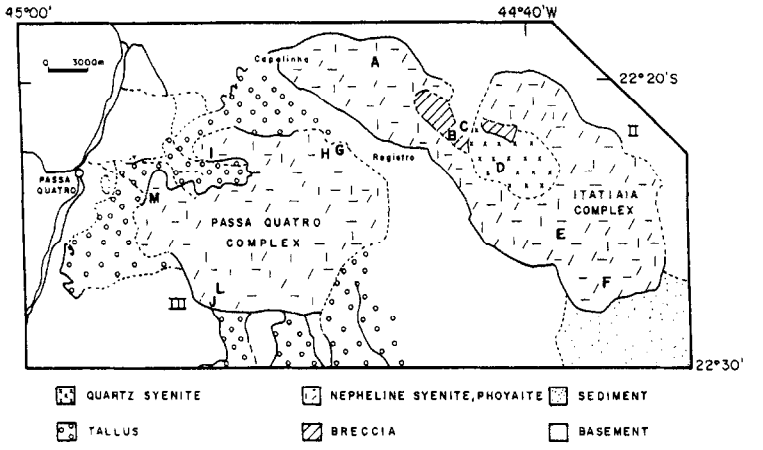


Fig. 2. Geological maps of the Passa Quatro and Itatiaia alkaline complexes (adapted from [15,16]). Letters *A* to *F* and *G* to *M* represent sampling areas from Itatiaia and Passa Quatro respectively.

geochronological data for Poços de Caldas are presented in Table 1 (see [10–14]). The K–Ar results suggest a long duration for magmatic emplacement of the complex (about 30 Ma). The whole-rock Rb–Sr age obtained on nepheline syenite throughout the region is around 88 Ma. The Rb–Sr and Ar–Ar ages corresponding to the last two datums of Table 1, however, indicate that hydrothermal and intrusive episodes occurred in a very short time span. Late local intrusions (i.e., lamproite and phonolite dykes) could account for the previous hypothesis of a long duration for the magmatic emplacement of the Poços

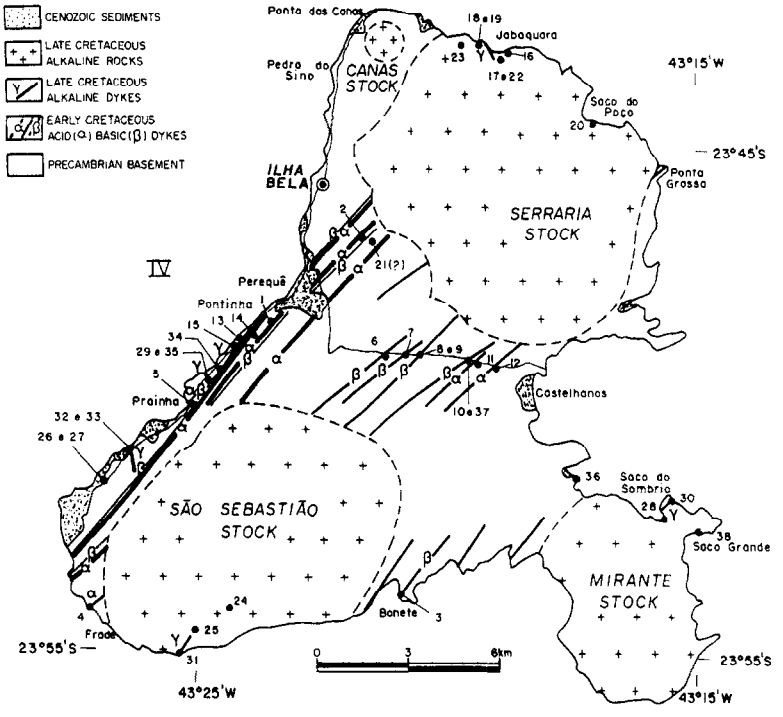


Fig. 3. Geological map of São Sebastião Island (modified from [23]). The numbers represent sampling sites.

de Caldas complex. As will be shown later, the paleomagnetic results support the hypothesis of a short time span for the emplacement of the main regional rocks from the complex.

2.2. Itatiaia complex

The Itatiaia alkaline complex (220 km^2, Fig. 2) is located on the boundary between the states of Rio de Janeiro and Minas Gerais. This intrudes into the Precambrian basement, which is represented by the Juiz de Fora metamorphic complex [7,17]. The Itatiaia massif is structurally linked to the SE Brazilian continental rift, which developed in an ENE to EW trending transcurrent system by the end of the Brasiliano (Pan African) Cycle and which was reactivated during the breakup of Gondwana [18]. The alkaline complex is composed of syenites, foyaites, pulaskites, quartz syenites, magmatic breccias and alkaline granite [19]. The mean K–Ar age obtained on ten biotites from the Itatiaia rocks is 70.5 ± 3.3 Ma [10,12].

2.3. Passa Quatro complex

The Passa Quatro alkaline complex (148 km^2, Fig. 2) is close to the Itatiaia complex. It was emplaced into the Precambrian metamorphic complexes of Juiz de Fora, Paraíba do Sul and the Açungui Group [7,17]. The main lithological types are alkali plutonic rocks (nepheline syenites) with minor phonolitic dykes. The K–Ar age obtained on

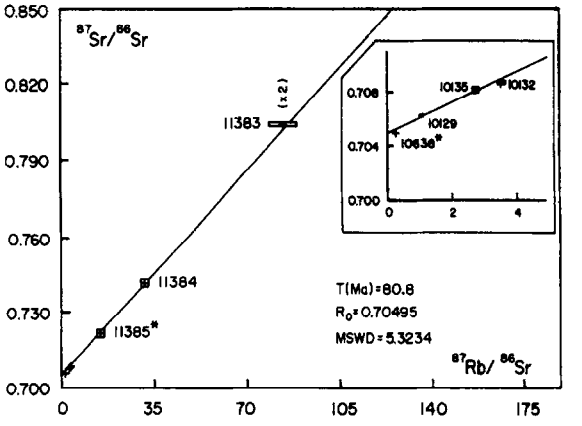


Fig. 4. Reference Rb–Sr isochron for alkaline rocks from São Sebastião Island. The asterisk indicates additional data (alkaline dykes) that were not considered for the isochron calculation.

only one sample of amphibole separated from a nepheline syenite is 67 ± 3 Ma [20]. The Rb–Sr whole-rock age calculated by Brotzu et al. [3] is 77 ± 3 Ma (R_0 = 0.70505), with MSWD = 6 [21]. This MSWD value indicates a perturbation of the Rb–Sr system [3]. The Rb–Sr age (whole rock) determined for this work was 70.4 ± 0.5 Ma (R_0 = 0.70540 ± 4). The initial Sr isotopic ratio in both determinations suggests that significant crustal contamination is not evident in these rocks.

Several factors, such as geomorphology, similar age and petrology, and the close relative location, suggest that the Passa Quatro and the Itatiaia complexes are comagmatic [19,22]. This possibility is discussed below using paleomagnetic results.

Table 2
Rb–Sr results for alkaline rocks from São Sebastião Island

S	NL	Rock type	Material	Rb (ppm)	Sr (ppm)	$^{87}Rb/^{86}Sr$	$^{87}Sr/^{86}Sr$
26	10129	syenite	WR	77.81	223.86	1.006±0.017	0.70615±5
28	10132	syenite	WR	150.68	124.20	3.512±0.059	0.70871±5
28	11383	syenite	biotite 1	405.54	7.12	168.0±4.7	0.90283±22
28	11384	syenite	biotite 2	204.77	18.87	31.53±0.62	0.74179±61
28	10135	syenite	feldspar	126.93	136.03	2.701±0.046	0.70820±6
32	10638	basanite	WR	65.67	856.36	0.222±0.004	0.70507±8
38*	11385	tinguaite	WR	271.17	59.47	13.22±0.33	0.72078±8

S = site number; NL = laboratory number; WR = whole rock; * = sample collected by Hennies and Hasui [26]. $^{87}Sr/^{86}Sr$ ratios were normalized to $^{86}Sr/^{88}Sr = 0.1194$; $\lambda(^{87}Rb) = 1.42 \times 10^{-11}$ yr^{-1}

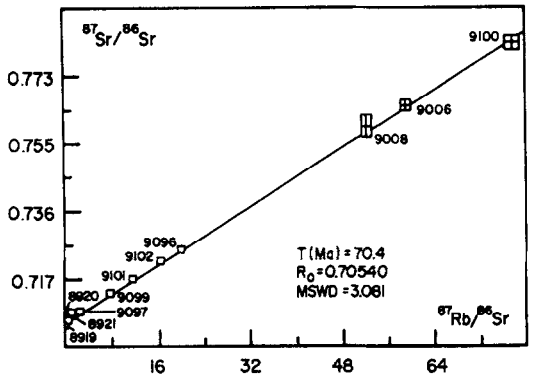


Fig. 5. Reference Rb–Sr isochron for nepheline syenite from the Passa Quatro complex.

2.4. São Sebastião Island

São Sebastião Island is located along the southeast coast of the state of São Paulo (Fig. 3). It is composed of Precambrian granitic rocks that were modified by the Brasiliano Cycle and it is intruded by Early Cretaceous subalkaline basic–acid dykes and by Late Cretaceous alkaline stocks and dykes ([24] and references therein). The Early Cretaceous dykes have a northeasterly trend and are of variable thickness (2–50 m) and extent (2–15 km). These dykes are cut by alkaline stocks (syenites). Chlorite and epidote were observed in most basic and acid dykes, indicating that the rocks suffered high-temperature alteration processes [24]. The Early Cretaceous acid–basic dykes can be considered as an easterly extension of the Santos–Rio de Janeiro swarm located inland [24]. A whole-rock K–Ar age determination for an acid dyke is 122.2 ± 2.8 Ma [25]. The alkaline stocks Serraria (65 km^2), São Sebastião (65 km^2), Mirante (38 km^2) and Canas (1 km^2) yielded a mean biotite K–Ar age of 84.0 ± 2.2 Ma [10,26]. The Rb–Sr age determined for this work was 80.8 ± 3.1 Ma, which is similar to K–Ar ages. The alkaline dykes are 0.5–3 m thick, and extend for up to a few hundred metres in a predominantly northeasterly direction. Their K–Ar ages (77.6 ± 7.8 Ma–82.8 ± 6.2 Ma) are similar to those of the alkaline stocks, but the dykes cut across these intrusions.

3. Experimental procedure and results

3.1. Fieldwork

At least three oriented hand samples were collected from every site for paleomagnetic work, using magnetic and sun compasses. From Passa Quatro and from the alkaline rocks of São Sebastião Island samples were collected for Rb–Sr age determinations.

From Poços de Caldas, 222 samples (73 sites) were collected for paleomagnetic work. Good outcrops along roads enabled a representative sampling of the complex (Fig. 1). For Itatiaia, 88 samples

Table 3
Rb–Sr results for nepheline syenites from the Passa Quatro Alkaline complex

Area	NL	Material	Rb (ppm)	Sr (ppm)	$^{87}Rb/^{86}Sr$	$^{87}Sr/^{86}Sr$
H	8921	WR	186.1	662.2	0.813±0.014	0.70654±8
I	8919	WR	138.9	1845.1	0.218±0.001	0.70549±5
I	8920	WR	151.1	1432.5	0.305±0.010	0.70564±10
L	9096	WR	142.4	20.8	19.83±0.33	0.72618±103
L	9097	WR	160.3**	61.7**	7.52±0.21	0.71313±7
M	9006	nepheline	203.4	10.1	58.79±0.90	0.7650±17
M	9008	WR	182.1	10.2	52.14±0.89	0.7598±30
M	9100	WR	197.2	7.4	77.45±1.37	0.7822±22
M	9101	WR	179.2	45.3	11.46±0.01	0.71680±8
M	9102	WR	171.4	30.5	16.28±0.18	0.72162±9
N	9099	WR	141.9**	173.4**	2.37±0.07	0.70789±7

** X-ray fluorescence analyses. For other notes, see Table 2

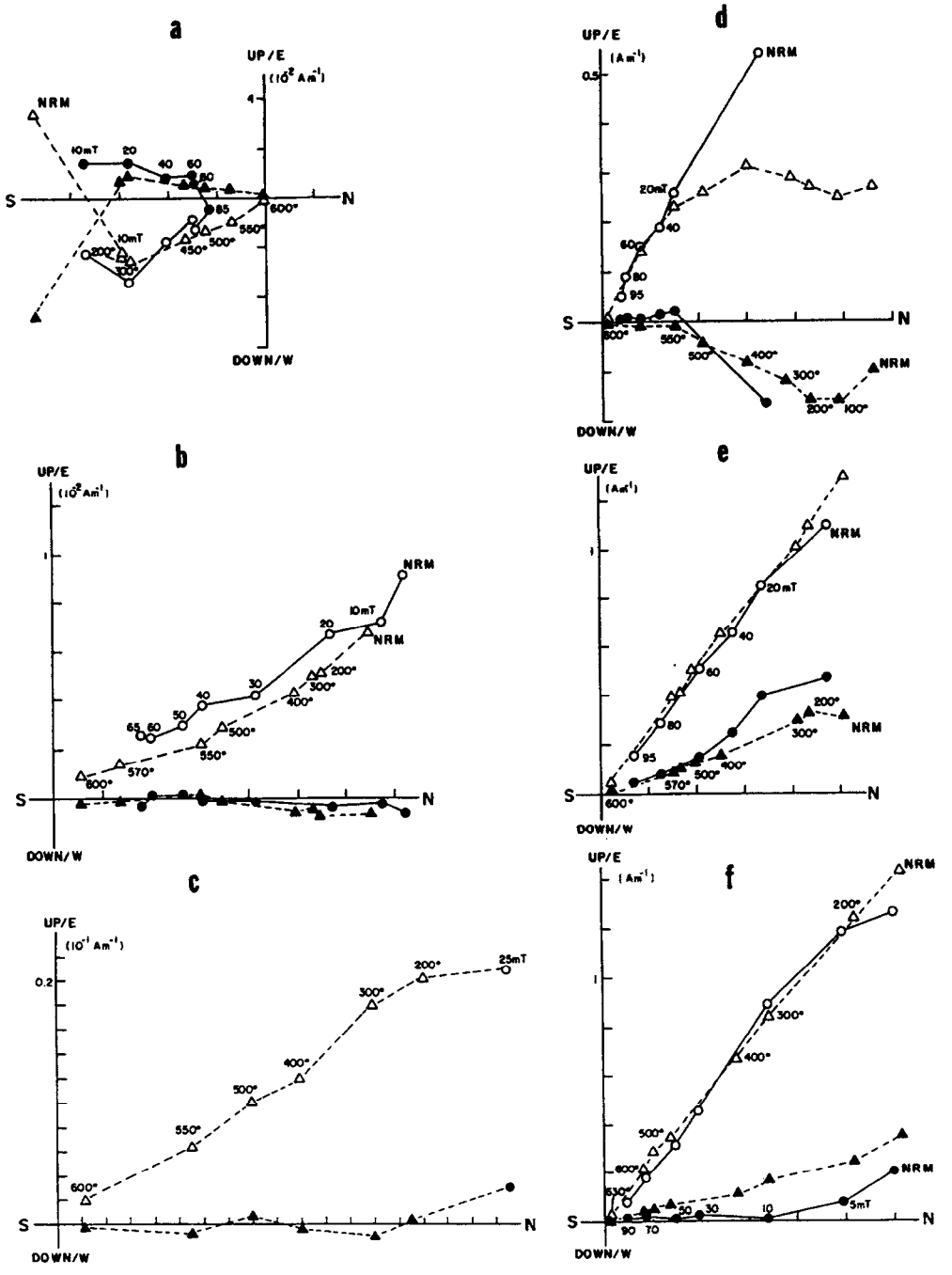


Fig. 6. Examples of Zijderveld diagrams for stable magnetization for Poços de Caldas (a), Itatiaia (b), Passa Quatro (c) and São Sebastião alkaline stocks (d), alkaline dykes (e) and basic dykes (f). Circles = AF demagnetization data points, with numbers indicating peak AF (mT). Triangles = progressive thermal demagnetization, with numbers indicating temperature (°C). Open symbols = vertical projections; closed symbols = horizontal projections.

Table 4
Paleomagnetic results for the Poços de Caldas alkaline complex

			Site mean direction					VGPs	
AREA	S	Nc/Nm	D(ChRM) (°)	I(ChRM) (°)	k	α_{95} (°)	R	LAT. (°S)	LONG. (°E)
A	6	3/3	190.4	37.4	32.5	21.9	2.9386	80.2	220.2
A	7	3/3	175.2	49.3	1006.2	3.8	2.9980	80.5	339.7
A	10	3/1	189.4	48.1	-	-	-	78.7	266.3
A	16	3/1	179.1	53.9	-	-	-	77.2	316.8
B	18	3/2	183.8	44.1	23951	1.6	2.0000	84.6	273.4
B	20	3/3	177.9	44.0	328.8	6.8	2.9939	85.5	338.9
B	21	2/1	192.6	44.1	-	-	-	77.7	245.2
B	22	3/3	179.7	43.7	39.6	19.8	2.9496	86.2	317.6
B	25	3/2	180.0	60.6	192.8	18.0	1.9948	70.2	313.4
B	26	2/1	172.1	47.0	-	-	-	80.8	357.9
B	27	3/1	189.1	44.7	-	-	-	80.5	253.8
B	28	3/2	180.3	33.9	189.9	18.2	1.9947	86.7	138.4
B	30	3/3	173.4	45.9	80.9	13.7	2.9753	81.8	359.8
C	31	3/2	194.4	36.8	254.9	15.7	1.9961	76.5	220.4
D	33	3/2	165.9	52.4	480.4	11.4	1.9979	73.3	358.9
D	34	3/2	162.4	55.7	139.7	21.2	1.9928	69.0	356.6
E	35	3/3	154.1	58.7	218.9	8.3	2.9909	61.9	359.2
E	36	3/3	187.3	47.4	707.7	4.6	2.9972	80.7	269.5
E	37	2/2	193.5	41.5	121.8	22.8	1.9918	77.4	234.4
E	38	3/3	184.4	39.5	912.0	4.0	2.9978	86.0	226.4
E	39	3/3	186.5	33.0	1062.2	3.7	2.9981	82.6	191.1
E	40	3/3	177.5	37.9	82.5	13.6	2.9758	87.5	61.0
E	41	2/1	179.2	46.0	-	-	-	84.6	321.1
E	42	3/3	177.8	45.2	163.9	9.6	2.9878	84.9	336.3
E	43	2/2	169.9	42.5	1240.2	7.0	1.9992	80.3	25.6
E	45	3/3	181.9	57.5	123.4	11.1	2.9838	73.7	308.0
F	46	3/3	181.8	48.5	64.6	15.4	2.9690	82.4	301.4
F	47	3/2	182.8	58.2	1297.6	6.9	1.9992	72.9	305.9
F	48	3/3	171.4	57.0	56.4	16.5	2.9645	72.7	337.1
G	49	7/4	11.4	-39.6	67.0	11.2	3.9553	79.4	228.4
H	50	3/1	4.9	-30.5	-	-	-	82.8	174.3
J	55	3/2	6.8	-51.2	260.6	15.5	1.9962	78.2	283.8
L	72	3/2	173.6	55.7	668.8	9.6	1.9985	74.5	333.2
M	57	8/4	174.9	49.2	404.2	4.5	3.9926	80.5	341.7
M	58	3/3	176.1	45.6	567.7	5.1	2.9965	83.7	347.2
M	59	2/2	156.1	54.3	157.5	20.0	1.9937	65.3	6.5
M	61	3/3	172.7	49.6	119.2	11.3	2.9832	79.1	349.3
N	62	3/1	181.9	37.7	-	-	-	88.1	200.9
N	63	3/1	198.4	33.1	-	-	-	72.3	214.3
N	64	3/3	166.9	44.3	84.1	13.5	2.9762	77.3	22.1
N	65	4/4	177.6	43.6	319.4	5.1	3.9906	85.8	345.0
O	66	3/2	199.5	54.2	153.1	20.3	1.9935	68.6	264.5
O	67	3/2	177.5	44.6	664.2	9.7	1.9985	85.1	341.1
O	68	3/3	173.4	53.6	706.4	4.6	2.9972	76.4	337.6
O	69	2/1	168.6	46.7	-	-	-	78.0	11.0
O	70	3/1	166.8	44.3	-	-	-	77.2	22.2
P	71	3/2	163.5	54.2	8116.5	2.7	1.9999	70.7	358.6

	Direction of magnetization				Pole			
MEAN 47 sites	D(°) 179.5	I(°) 47.0	k 64.1	α_{95}(°) 2.6	LAT(°S) 83.2	LONG(°E) 320.1	K_{SC} 2609.8	A_{95}(°) 2.7

S = site number; N_c/N_m = number of collected samples per site/number of samples used for mean determinations; D(ChRM), I(ChRM) = declination and inclination of the characteristic remanent magnetization; k, α_{95} = Fisher parameters [31]; R = resultant vector; K_{SC}, A_{95} = McFadden parameters [32]

were collected from 28 sites (Fig. 2). The rocks were usually more strongly weathered than those from Poços de Caldas. From Passa Quatro, 31 samples (9 sites) were collected for paleomagnetism and 6 samples were taken for geochronology, but difficult access and weathered outcrops prevented extensive sampling (Fig. 2). Alkaline and basic–acid rocks were collected from São Sebastião Island because the Early Cretaceous basic–acid dykes could have been remagnetized by the Late Cretaceous alkaline stocks. Because of difficult access the sampling of alkaline stocks was restricted to their peripheries: 75 samples (25 sites) were collected, 48 from alkaline rocks (stocks and dykes, 14 sites) and 27 from basic–acid rocks (dykes, 11 sites) (Fig. 3).

3.2. Radiometric ages

Rb–Sr isotopic analyses were carried out using standard procedures at the *Centro de Pesquisas Geocronológicas—USP* (São Paulo). Both whole-rock and mineral-separate samples were analyzed. The isotopic measurements were performed on a VG 354 multicollector and with Varian TH-5 mass spectrometers. Most Rb and Sr determinations were done by the isotope dilution method (uncertainties $< 1\%$). The reliability of Rb–Sr isochrons was evaluated from the the precision of the analytical data and through MSWD analysis. MSWD values were evaluated considering the number of determinations used for the isochron [27]. Because the MSWD values obtained for Passa Quatro and São Sebastião Island were higher than expected, considering the number of data, the analytical errors for the ages and the Sr initial ratios were recalculated following Kawashita et al. [28].

The Rb–Sr elemental and isotopic values obtained for alkaline rocks from São Sebastião Island are shown in Table 2. Using Williamson's isochron modeling [29], an age of 80.8 ± 3.1 Ma ($R_0 = 0.70495 \pm 13$) was obtained for the alkaline stocks (Fig. 4). In spite of the high MSWD value (5.32), the Rb–Sr age is in agreement with the previous K–Ar ages estimated at 84.0 ± 2.2 Ma [10,26], which suggests that this age probably represents the time of emplacement of the syenitic stocks. It is interesting

Table 5
Paleomagnetic results for the Itatiaia complex

			Site mean direction					VGPs	
AREA	S	Nc/Nm	D(ChRM) (°)	I(ChRM) (°)	k	α_{95} (°)	R	LAT. (°S)	LONG. (°E)
B	7	3/3	340.7	-39.7	109.5	11.8	2.9817	72.1	40.9
B	8	3/3	328.0	-41.5	40.7	19.6	2.9508	60.6	36.0
B	9	3/3	7.6	-58.9	104.7	12.1	2.9809	71.5	296.5
B	10	3/2	350.1	-43.4	79.3	28.4	1.9874	80.5	25.4
B	11	3/3	5.9	-64.9	52.6	17.1	2.9620	65.0	305.7
C	12	3/2	7.6	-30.8	80.8	28.2	1.9876	80.8	187.8
C	13	3/2	340.0	-42.5	476.3	11.5	1.9979	71.5	34.3
C	14	3/2	356.6	-23.1	1171.5	7.3	1.9991	79.1	117.2
D	16	3/3	344.2	-50.2	501.4	5.5	2.9960	73.5	10.6
D	17	3/3	355.9	-49.5	2032.1	2.7	2.9990	81.2	339.2
D	18	3/2	3.2	-63.2	295.5	14.6	1.9966	67.5	309.4
E	19	3/2	161.0	54.0	205.1	17.5	1.9951	69.4	5.2
E	24	3/2	6.1	-59.5	463.7	11.6	1.9978	71.4	300.7

	Direction of magnetization				Pole			
MEAN	D(°)	I(°)	k	α_{95}(°)	LAT(°S)	LONG(°E)	k	α_{95}(°)
13 sites	351.8	-48.5	28.5	7.9	79.1	353.2	27.5	8.1

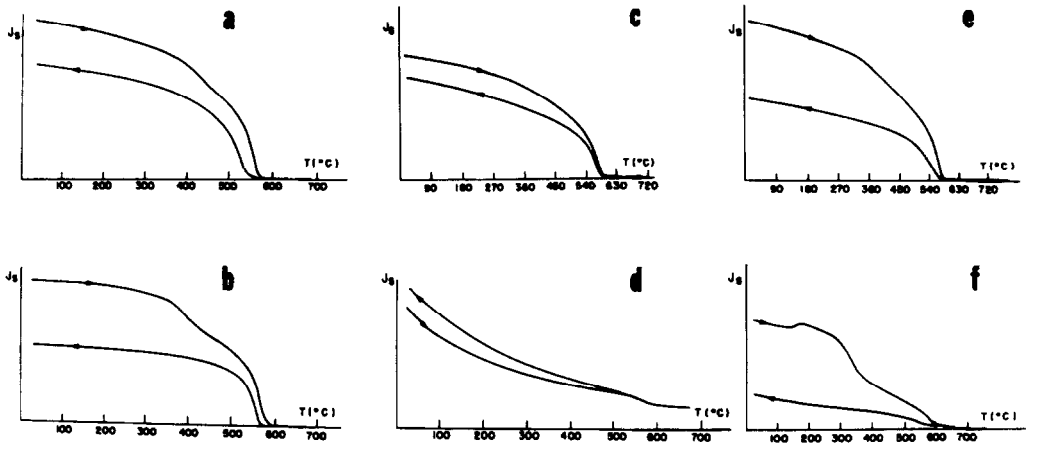


Fig. 7. Thermomagnetic curves for alkaline (a–d) and basic–acid rocks (e,f).

to observe that although the alkaline dykes cut across the syenitic stocks their ages might be similar (Fig. 4).

In Table 3 the Rb–Sr elemental and isotope values determined for the Passa Quatro complex are presented. The reference Rb–Sr isochron diagram for the nepheline syenites (Fig. 5) shows an age of 70.4 ± 0.5 Ma ($R_0 = 0.70540 \pm 4$). This value is lower than that presented by Brotzu et al. [3]. The calculated MSWD (3.08) suggests a possible perturbation of the Rb–Sr system as suggested by Brotzu et al. [3]. The Rb–Sr age agrees with the K–Ar age [20] and shows that the complex was emplaced during the most important period of alkaline magmatic activity.

3.3. Paleomagnetism

Remanent magnetizations

At least four 2.5 cm diameter cores were cut from every oriented block, and from these oriented cores at least three 2.5 cm high cylindrical specimens were cut and used for paleomagnetic measurements. Molspin MS1 and Schoenstedt DSM-2 spinner magnetometers were used for the measurements. The thermal and alternating field (AF) demagnetization pro-

Table 6
Paleomagnetic results for the passa quatro alkaline complex

			Site mean direction					VGPs	
AREA	S	Nc/Nm	D(ChRM) (°)	I(ChRM) (°)	k	α_{95} (°)	R	LAT. (°S)	LONG. (°E)
G	29	3/2	355.1	-56.1	176.7	18.8	1.9943	75.0	330.6
H	30	3/3	354.1	-36.0	145.9	10.2	2.9863	83.9	67.7
H	31	3/3	338.9	-41.3	25.2	25.0	2.9207	70.5	37.1
I	32	3/3	166.6	43.7	808.9	4.3	2.9975	77.3	28.1
M	37	6/4	355.1	-44.7	291.6	5.4	3.9897	84.1	2.8

	Direction of magnetization				Pole			
MEAN	D(°)	I(°)	k	α_{95}(°)	LAT(°S)	LONG(°E)	k	α_{95}(°)
5 sites	349.7	-44.5	81.6	8.5	79.8	18.7	82.3	8.5

cedures were, in general, similar for all samples. At least three specimens per sample were subjected to demagnetization treatment.

One pilot specimen from every sample was subjected to detailed AF treatment in 5 mT steps up to, in general, 80 mT. In some cases, when the magnetization components had high coercivity, fields up to 150 mT were applied. Another pilot specimen was subjected to detailed thermal demagnetization from 200°C to, in general, 650°C, in steps of 50°C up to 500°C; above this temperature steps of 20 or 30°C were employed. The other specimens were subjected to at least six temperature or AF steps selected with the pilot specimens. Changes in magnetic susceptibility during thermal treatment were monitored with Bison 3101A and Sapphire S1 instruments. For most samples from São Sebastião Island, the optimum temperatures and/or AFs were, respectively, between 450 and 530°C and 30 and 40 mT. For almost all samples from Passa Quatro and Itatiaia the optimum temperatures were between 300 and 500°C. For Poços de Caldas the optimum temperatures were

Table 7
Paleomagnetic results for São Sebastião Island

		Site mean direction					VGPs	
S	Nc/Nm	D(ChRM) (°)	I(ChRM) (°)	k	α_{95} (°)	R	LAT. (°S)	LONG. (°E)
1	3/3	345.2	-47.8	194.5	8.9	2.9897	75.7	20.4
7	3/3	9.2	-58.7	147.4	10.2	2.9864	72.5	290.6
14	4/3	6.0	-52.2	199.3	8.8	2.9900	79.5	285.9
5	4/2	343.8	-60.6	101.0	25.1	1.9901	67.8	348.0
6	4/3	347.1	-54.7	110.7	11.8	2.9819	74.1	356.4
2*	3/3	15.2	-49.7	242.8	7.9	2.9918	74.9	254.2
3*	3/3	2.9	-49.6	151.3	10.1	2.9868	82.9	293.6
4*	3/3	358.9	-52.6	251.1	7.8	2.9920	80.6	320.4
15*	3/3	333.4	-57.1	198.2	8.8	2.9899	63.4	7.3
18*	3/3	341.7	-51.7	43.4	18.9	2.9539	71.8	12.8
20*	3/3	10.9	-59.6	64.4	15.5	2.9690	71.1	288.2
10**	3/3	355.9	-45.1	283.4	7.3	2.9929	85.3	6.1
11**	3/2	9.6	-46.4	377.2	12.9	1.9973	80.5	250.9
12**	3/2	355.1	-35.4	156.3	20.1	1.9936	83.7	86.4
13**	3/1	338.3	-59.2	-	-	-	65.7	357.9
19**	3/3	5.8	-50.9	256.2	7.7	2.9922	80.8	282.2
21**	3/3	359.0	-58.9	114.7	11.6	2.9826	74.1	317.4
23**	2/2	351.7	-51.2	245.5	16.0	1.9959	79.1	355.3
25**	3/2	163.3	71.6	89.1	26.8	1.9888	55.3	330.9

Alkaline Rocks

	Direction of magnetization				Pole			
MEAN	D(°)	I(°)	k	α_{95}(°)	LAT(°S)	LONG(°E)	k	α_{95}(°)
11 sites	356.0	-54.8	77.1	5.2	77.8	330.6	43.9	7.0

Basic-acid Dykes

	Direction of magnetization				Pole			
MEAN	D(°)	I(°)	k	α_{95}(°)	LAT(°S)	LONG(°E)	k	α_{95}(°)
7 sites	357.0	-49.9	61.3	7.8	82.0	334.8	52.9	8.4

1 = alkaline stocks; 2* = alkaline dykes; 10** = basic–acid dykes. Site 25 was not used in the mean

between 400 and 550°C. Viscous remanent magnetization components were observed for most samples but could generally be eliminated with a 10–20 mT AF treatment. Examples of stable magnetization behaviour for the rocks studied can be seen in Fig. 6.

The characteristic remanent magnetization (ChRM) per specimen was determined through the best grouping of directions obtained after several demagnetization steps, together with the corresponding Zijderveld diagram [30]. The ChRM per sample was calculated with Fisherian statistics [31] from the mean ChRM using all specimen values. Mean sample directions were then used to calculate site mean directions and the corresponding virtual geomagnetic poles (VGPs).

The paleomagnetic results for Poços de Caldas, Itatiaia, Passa Quatro and São Sebastião Island are shown in Tables 4 and 5, 6 and 7 respectively. Only sites with stable magnetization directions are represented. For several sites, the magnetization direction

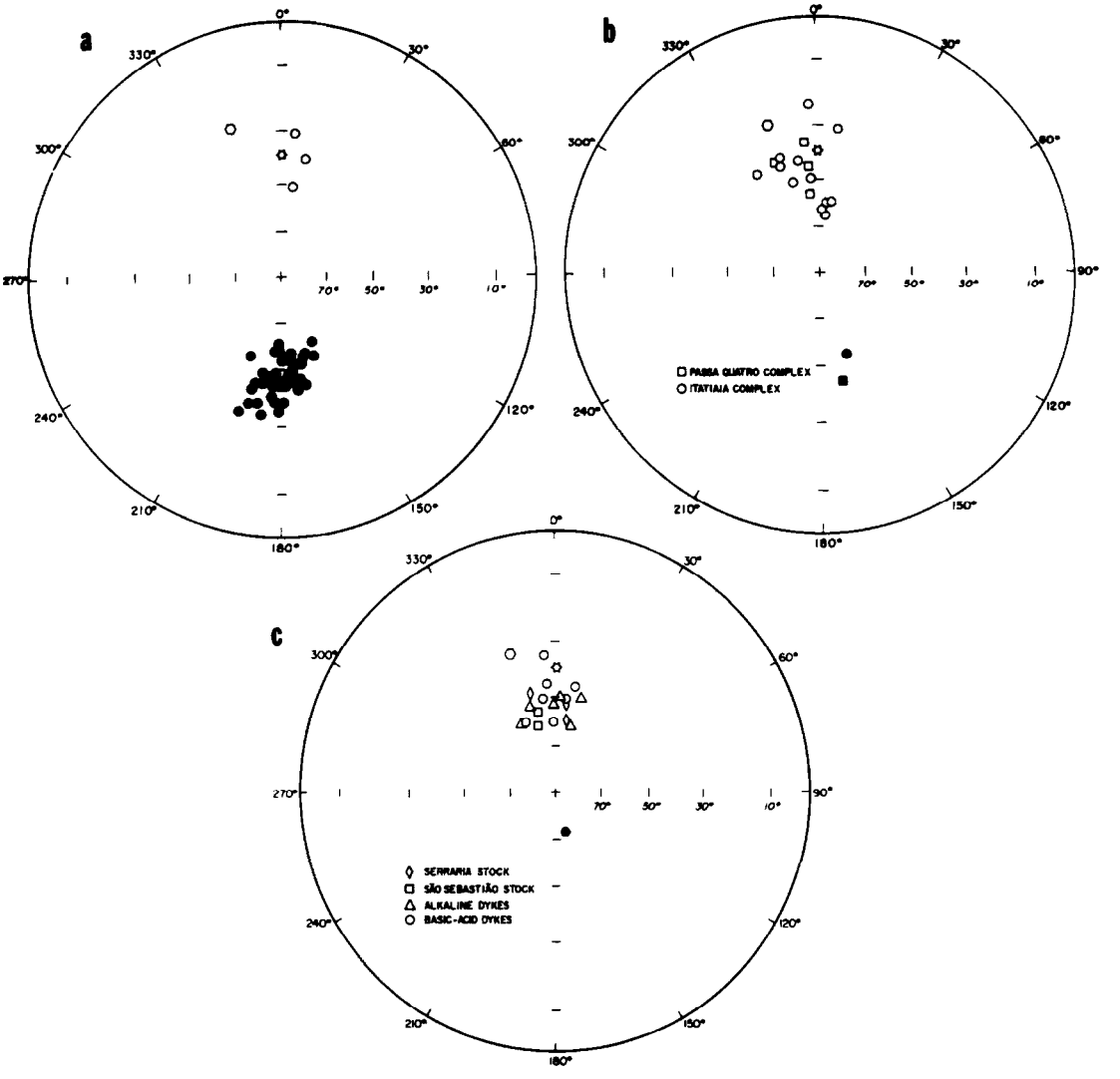


Fig. 8. Site-mean directions for Poços de Caldas (a), Passa Quatro and Itatiaia (b) and for São Sebastião alkaline and basic–acid rocks (c). Open and closed symbols = normal and reverse polarity respectively. Hexagon and star = geomagnetic field and dipolar field directions (equal-angle projection).

was obtained with only two samples and, although in these cases the statistical quality of the means cannot be estimated from the calculated α_{95} values, they have been included because in general the directions are coherent with the whole dataset.

For the Poços de Caldas alkaline complex, 57% (126) of the samples yielded consistent paleomagnetic results. Thermal treatments were, in general, more effective than AF treatments in eliminating secondary magnetization components for these rocks. Normal polarity was observed for ankaratritic rocks whereas the other lithologies presented reverse polarity. Of the 88 Itatiaia alkaline samples, 56 (64%) had to be discarded because it was impossible to find stable directions. Thermal treatments were better than AF treatments. Almost all the samples presented normal polarity. Consistent paleomagnetic results were obtained for 15 (48%) of the samples from the Passa Quatro complex. The samples were, in general, subjected to thermal treatment and normal polarity was observed. Samples from alkaline and basic–acid dykes of São Sebastião Island showed high

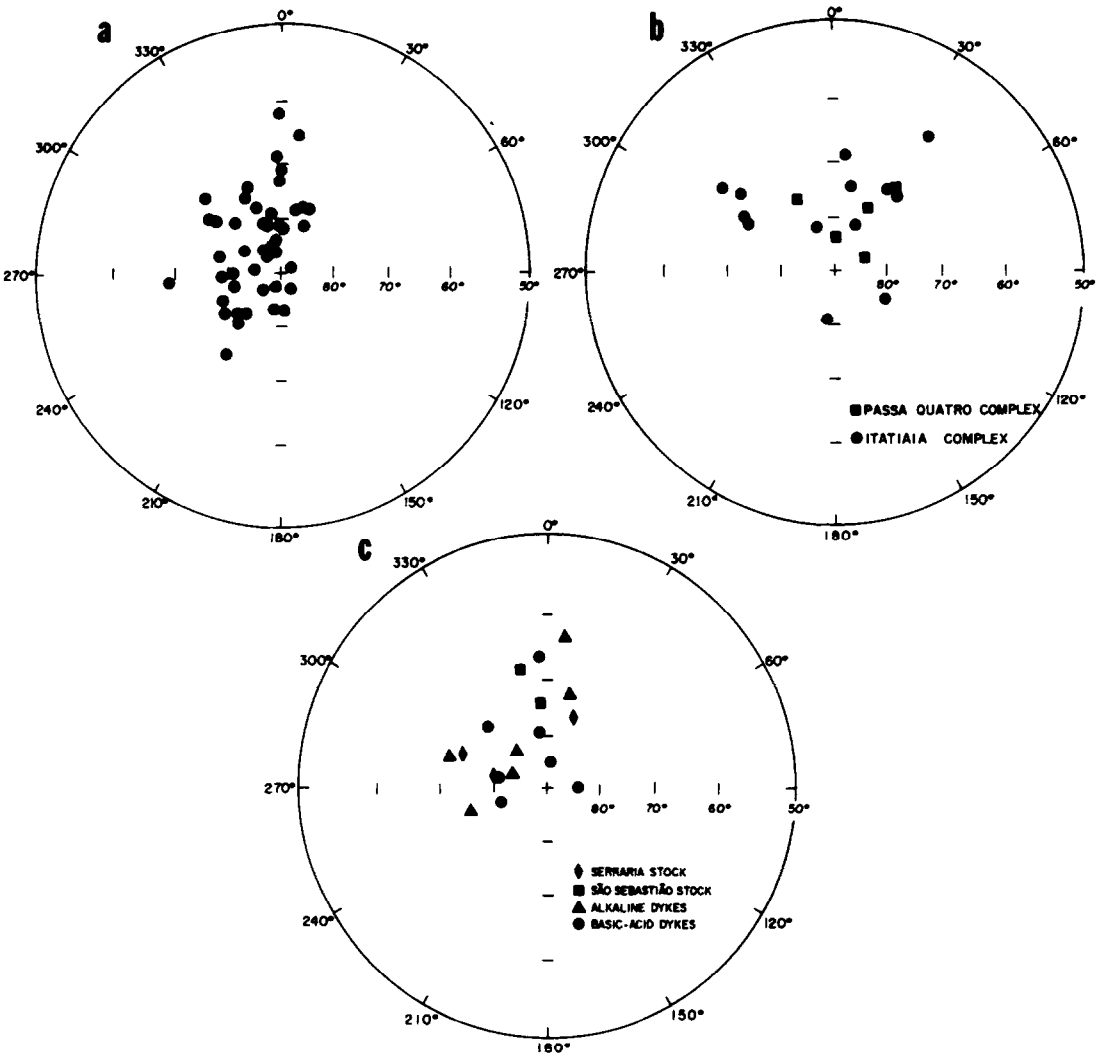


Fig. 9. Southern geomagnetic virtual poles for Poços de Caldas (a), Passa Quatro and Itatiaia (b) and São Sebastião (c) (equal-angle projections).

Table 8
Angular dispersion of VGPs for the Poços de Caldas (CL), Passa Quatro + Itatiaia (PQ + IT) and São Sebastião Island (ISLAND) datasets

Data Set	No. of VGPs	S (°)	S_u (°)	S_l (°)	k_B
CL	47	9.9	11.6	8.6	67.2
PQ+IT	18	12.8	16.8	10.3	40.3
ISLAND	18	10.5	13.8	8.5	59.5

S = angular dispersion [34]; S_u,S_l = upper and lower 95% confidence limits for S [35]; k_B = between-site precision parameter [32]. The latter was used to determine the angular dispersion S

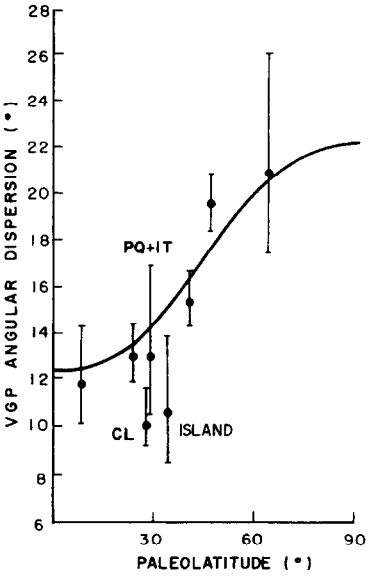


Fig. 10. Virtual geomagnetic pole dispersions observed for Poços de Caldas (CL), Passa Quatro + Itatiaia (PQ + IT) and São Sebastião (ISLAND) compared with the global compilation of McFadden and McElhinney [36] for the 45–110 Ma period.

magnetic stability. For 67% of the samples (i.e., 14 samples) from the alkaline dykes and 75% (9 basic and 9 acid) of samples of the basic–acid dykes, consistent paleomagnetic results could be determined. For 27 samples of the alkaline stocks, 14 (52%) presented coherent paleomagnetic results. Normal polarity was observed for all analyzed samples except for those from an acid dyke, but in this case the magnetic inclination is not compatible with the dataset.

Thermomagnetic analysis

Alkaline and basic–acid rock samples from all the studied units were analyzed with a thermomagnetic balance for the determination of Curie temperatures (T_C) and the identification of magnetic minerals. For all alkaline rocks the thermomagnetic curves were similar and, in general, two magnetic phases were observed. The first one with transition temperatures between 350 and 450°C probably indicates the presence of maghemite, whereas the second with Curie temperatures between 570 and 594°C is characteristic of titanomagnetites with a composition close to that of magnetite (Fig. 7a–c). For tinguaites from the Poços de Caldas complex, the Curie temperatures were not well defined (Fig. 7d) due to superposition of paramagnetic effects from the matrix on the ferrimagnetic mineral signal. Highly oxidized titanomagnetites (T_C = 576–594°C) and maghemites (367–380°C) were observed for basic–acid rocks from Early Cretaceous dykes from São Sebastião Island (Fig. 7e,f). Kink features were observed in two basic dykes suggesting that these rocks suffered hydrothermal alteration processes ([33]; Fig. 7f).

Table 9
Comparison of VGPs for the Passa Quatro (1) and Itatiaia (2) alkaline complexes

N_1	N_2	R_1	R_2	R	k_1	k_2	A	B	(1-p)x100%
5	13	4.95138	12.56320	17.50270	82.27590	27.47370	0.03905	0.20591	45.8

N = number of VGPs; R_1,R_2 = resultant vector of 1 and 2; R = sum vector $R_1 + R_2$; k_1,k_2 = Fisher parameters [31]. $A = r[(R_1 + R_2)^2 - R^2]/2[(N_1 - R_1) + r(N_2 - R_2)](R_1 + rR_2)$; $p = 0.05$; $r = k_2/k_1$ $B = (1/p)^{1/N-2} - 1$; $p = 0.05$; $N = N_1 + N_2$ The last column gives the probability of distinctness between the VGP groups

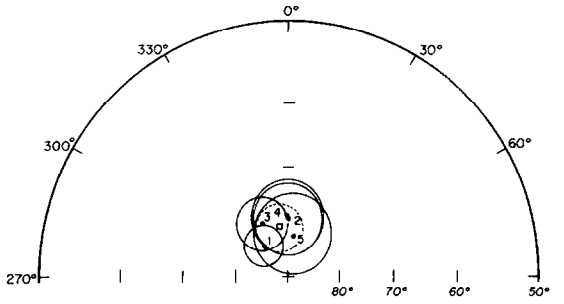


Fig. 11. Equal-angle projections of paleomagnetic poles for Late Cretaceous rocks of (*1*) Poços de Caldas, (*2*) Passa Quatro and Itatiaia, (*3*) São Sebastião, (*4*) Patagonia basalts [37] and (*5*) Patagonia sedimentary rocks [38]. Square = mean paleomagnetic pole; circles surrounding poles = 95% confidence limits.

Characteristic remanent magnetizations (ChRM), virtual geomagnetic poles (VGPs) and statistical tests

The site-mean directions of the ChRM for all units are presented in Fig. 8 (Tables 4–7). Fig. 9 shows the VGPs calculated for every sampling site. As can be seen in Fig. 8a three sites from Poços de Caldas exhibit normal polarity while all the other sites show reverse polarity. The normal polarity sites correspond to ankaratritic rocks interpreted as representing the earliest magmatic activity [8,10,11,9]. A paleomagnetic pole was determined by calculating the average of the 47 VGPs (Table 4). To evaluate whether geomagnetic secular variation has been adequately sampled, angular dispersion S [34] of the VGPs was calculated (Table 8) and compared with the global compilation of paleosecular variation data presented by McFadden and McElhinny [36] for the 45–110 Ma age range (Fig. 10). Fig. 10 shows that VGP angular dispersion for the Poços de Caldas dataset is lower than expected from the global compilation. This could indicate that the duration of emplacement for the Poços de Caldas complex was shorter than required for adequate sampling of paleosecular variation. Nevertheless the paleomagnetic pole and its 95% confidence circle are consistent with other Late Cretaceous poles (Fig. 11). The paleomagnetic pole obtained in this work does not accord with that reported earlier for Poços de Caldas [39]. It is probable that the difference reflects the smaller number of sampling sites and the use of only AF treatment in the earlier work.

As mentioned earlier, several factors suggest that the Passa Quatro and the Itatiaia complexes are comagmatic [19,22]. This possibility was investigated with paleomagnetic data through the statistical procedure of McFadden and Lowes [40]. This procedure was applied to estimate whether VGPs obtained for the two complexes represent distinct directions at the 95% confidence level (Table 9). The results indicate that the two sets of VGPs cannot be considered distinct at 95% confidence. Although the relatively small number of available data for Passa Quatro and Itatiaia presents major limitations such statistical tests are consistent with the hypothesis that the complexes are comagmatic. Therefore a single paleomagnetic pole has been calculated for Passa Quatro and Itatiaia (mean direction $N = 18$, $D = 351.2°$, $I = -47.4°$, $k = 35.4$, $\alpha_{95} = 5.9°$; pole $N = 18$, 360.0°E, 79.5°S, $K_{SC} = 607.4$, $A_{95} = 5.7°$). The VGP angular dispersion [34] for the Passa Quatro and Itatiaia dataset (Table 8) is compatible with the global compilation of paleosecular variation for the 45–110 Ma interval ([36]; Fig. 10).

As may be seen from Fig. 8c, the directions of ChRM for basic–acid dykes (Early Cretaceous) and for alkaline rocks (Late Cretaceous) from São Sebastião Island samples are similar. This observation suggests that the Early Cretaceous basic–acid dykes may have been remagnetized by the intrusion of alkaline stocks. The statistical results reported in Table 10 [40] are consistent with this suggestion. Although the small number of available datasets

Table 10
Comparisons of VGPs of alkaline rocks (Late Cretaceous, 1) and basic–acid rocks (Early Cretaceous, 2)

N_1	N_2	R_1	R_2	R	k_1	k_2	A	B	(1-p)x100%
11	7	10.87025	6.902065	17.75713	77.06888	61.265	0.06315	0.20591	62.5

For notes, see Table 9

limits the reliability of this analysis, remagnetization can be expected in view of the large volume of alkaline stocks that cut across the Early Cretaceous dykes. The paleomagnetic analysis of Early Cretaceous dykes from the Santos–Rio de Janeiro swarm has shown that units of both polarities are present and the magnetization directions are rather different from those of dykes of the same age from São Sebatião Island (M. Ernesto, pers. commun.). Another indication that the basic–acid dykes were affected by alkaline stocks comes from petrographic observations of the dykes that reveal effects of high-temperature alteration [24]. Considering all the evidence, a single paleomagnetic pole was calculated that includes the data for Early Cretaceous dykes and Late Cretaceous alkaline rocks (called the ISLAND set) from São Sebastião Island (mean direction $N = 18$, $D = 356.4°$, $I = -52.9°$, $k = 70.0$, $\alpha_{95} = 4.2°$; pole $N = 18$, 331.9°E, 79.4°S, $K_{SC} = 862.4$, $A_{95} = 4.9°$). The upper limit of the VGP angular dispersion [34] for the ISLAND set (Table 8) is just slightly lower than that expected from the global compilation for the 45–110 Ma period([36]; Fig. 10).

4. Discussion and conclusions

The paleomagnetic poles obtained for the studied alkaline complexes are represented in Fig. 11. Their 95% confidence limits intersect and the poles also agree with the pole from Patagonia basalts which was considered as representative for the Late Cretaceous ([37]; Fig. 11). Recently Somoza and Vilas [38] obtained a paleomagnetic pole from sedimentary sequences in Patagonia that also agrees with these determinations (Fig. 11). By considering that (a) the poles calculated for the ISLAND dataset, Passa Quatro + Itatiaia (PQ + IT), Poços de Caldas (CL), the Patagonia basalts and Patagonia sedimentary rocks present a reasonable number of data, (b) the pole positions are internally consistent, (c) the geomagnetic secular variation has probably been adequately averaged at least for two of the poles and (d) that the 95% confidence circles are small, a mean paleomagnetic pole was calculated for the Late Cretaceous from these five poles (349.1°E, 81.0°S, $\alpha_{95} = 3.5°$, $k = 492.2$). This mean paleomagnetic pole is considered to be a reliable constraint for the Late Cretaceous South American APWP and consequently useful for tectonic interpretations.

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